

APPLICATION

The R401 control is a highly configurable microprocessor-based heat pump controller that is simple to use, and yet provides all the necessary features to improve the operation and safety of water source heat pumps.

General Operating Parameters

Operating Environment: -20°F to 140°F and up to 90% relative humidity, non-condensing.

Storage Environment: -30°F to 150°F and up to 90% relative humidity, non-condensing.

Power Requirements:

R401 only- power draw -

Nominal 5 VA draw at 24VAC.

Maximum 9 VA draw at 24VAC.

A dedicated 24VAC, 50-60Hz, 1Ph, 40VA transformer minimum is required for a typical WSHP application.

Relay Contact Ratings

The following relays are mounted on the R401 control:

Compressor Relay: 40VA at 24VAC.

Alarm Relay: 28VA at 24VAC.

Fan Relays (3) 28VA at 24VAC

Accessory Relays (2) 28VA at 24VAC

Auxiliary/Emergency heat Relay (1) 28VA at 24VAC

Grounding

The control board is grounded through two metal standoffs.

Product Specification

Features

- Anti-short cycle protection
- High- and Low-pressure cutouts
- Water Coil low temperature cut-out
- Air Coil low temperature cut-out
- Random Start
- Unit Performance Sentinel
- Over/Under Voltage protection
- Diagnostic LED
- Reset Lockout at unit or disconnect
- Intelligent Reset
- Condensate Overflow sensor
- Test Mode
- Electric Heat Outputs
- Accessory Water Valve Connection
- Optional delays
- DC Fan speed staging
- Vacated premises Pump Circulation timer (Optional add)
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Board Switch Options

DIP SWITCH 1 (Off/On values)

1.1 Compressor Start Delay (0/ 5 sec)

1.2 Motor Type (PSC / DC)

1.3 Blower Time Delay (0/45 seconds)

1.4 Dehumidification (Normal/Dehumidification mode- applies only in Cool mode)

1.5 Boilerless Operation (Normal/ Boilerless operation)

1.6 Boilerless Changeover Temp. (40F/50F)

1.7 Vacated Premises Timing (four 15 min cycles per day/ eight 15 min cycles per day- requires vacated Premises kit)

1.8 Not used

DIP SWITCH 2 (Off/On values)

2.1 Accessory Relay 1 Activation (Disabled/Enabled)

2.2 Accessory Relay 1 - Pre-Compressor Initialize (0/60 seconds)

2.3 Accessory Relay 2 Activation (Disabled/Enabled)

2.4 Accessory Relay 2 - Post Fan Delay (0/30 seconds)

Notes about dipswitches:

- a. All dipswitches come from factory in OFF position
- b. Before resetting dipswitches cut off power

Jumper Selections: (Not cut/cut values)

JW1- Water Coil Temperature cutout (30F/10F)

JW2- Air Coil Temperature cutout (30F/10F)

JW3- Alarm Relay output (AL1 -24Vac on alarm/ AL1 dry contact with AL2 on alarm)

Indicators and Safety features:

LED: ON green (normal mode); OFF Micro stopped-board non functional;
ON yellow (board in test mode)

Delays: Random Start (5-80 sec), 5 minute anti cycle compressor delay between compressor starts and at initial power up.

Display:

The 2-digit display code indicates Normal and Abnormal operation (fault codes) as follows:

Normal Modes of operation:

St - Standby (no outputs)

FO - Fan only operating

CO – Cooling Mode (if flashing, compressor is in anti cycle delay)

H1 – Heating Mode (if flashing, compressor is in anti cycle delay)

A1- Accessory relay 1 Activated

A2 - Accessory relay 2 activated

Fault Codes:

Abnormal Conditions (* indicates that the fault code will be preceded by “RE” -retry, with three attempts to see if the condition has returned to normal. On the third try if condition is still abnormal, it will be preceded by LO-lockout, and the compressor will be locked out until reset by powering down)

12 High Pressure*

13 Low Pressure *

14 Water Coil Low Temperature Cut Out *

15 Air Coil Low Temperature Cutout *

16 Condensate Overflow*

17 Over/under Voltage (Auto resets when back to normal)

18 (In Test mode Only) Water Coil and Air Coil sensors are reversed

19 Water Coil and/or Air Coil Sensor shorted or open*

Ay Optional Vacated Premises Function enabled but thermostat was not left in COOL mode

Test mode:

Installer can enter Test Mode by jumping test pins momentarily. In test mode all time delays are sped up 15 times. Yellow LED will be ON and display will show last fault stored in memory (a display of 11 means there is no fault stored in memory)

Outputs:

Alarm Relay (activated upon lockout) Dry contact or 24Vac depending on JW3 jumper

A1 and A2 Accessory Relays (operate according to dipswitch 2 settings)

CC/CCG- Compressor Relay Output

W1 Relay – Aux heat (Pass through from thermostat)

Terminal ‘A’ follows compressor activation for water valve opening applications

Reversing Valve Relay -Pass through from thermostat Note: 24V from thermostat on O/B terminal will energize relay AND turn board into Cool Mode

Fan relays 1, 2, 3- Fan speed Relays-see below

Motor Fan Speeds:

When a DC Motor is specified via Dipswitch 1.2, the fan Speed Relays behave as follows:

G signal only – activate Fan relay 1

Y+O+ G signals (Cool Mode)– activate Fan relay2; after 10 minutes activate Fan relay 3*

Y signal only (Heat Mode) – activate Fan relay 3

* Skip if Dipswitch 1.4 ON

When a PSC Motor is specified via Dipswitch 1.2, the fan Speed Relays behave as follows:

G signal only – activate Fan relay 1

Y+O+ G signals (Cool Mode)– activate Fan relay1; after 10 minutes activate Fan relay 2*

Y signal only (Heat Mode) – activate Fan relay 2

* Skip if Dipswitch 1.4 ON

Special Operating Modes:

Boilerless Operation:

In boilerless mode the compressor is used for heating mode only when CO1 (Water Coil temp) is above the temperature set by dip switch 1.6. If below this temperature, the compressor is de-energized. The compressor will be locked out for 60 minutes to prevent nuisance cycling.

Operational Flowchart

Initial Condition:

Fan: Off

Compressor: Off

Away Option Chip: Not present

